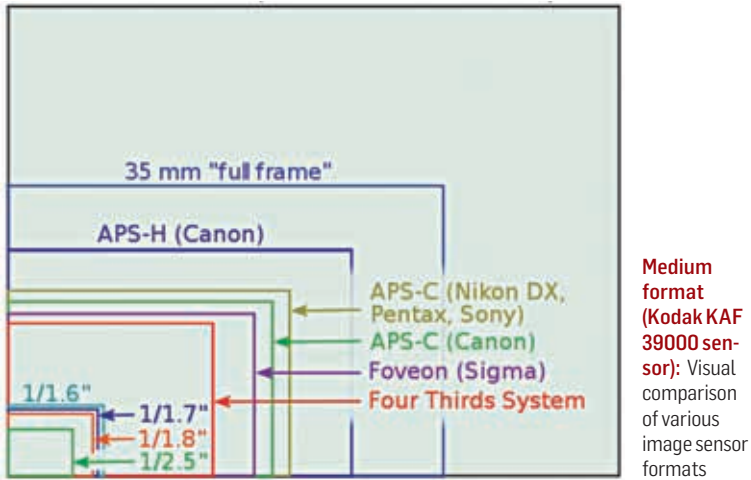




large area for contamination per component leads to a further sizeable reduction in yield. Moreover, a full-frame sensor in its photolithography stage needs three different exposures, each requiring separate masks and quality control steps. On the other hand, with a single mask the largest size possible is the APS-H, making it a common option to keep production costs under control and manage better sufficient yields. Slight variations within a nominal format are common as semi-conductor fabrication and processing are a continuously evolving domain. Also, third-party foundries are often used to source in sensors.

Crop factor

Since most digital camera sensors are smaller than film, the image as seen from these cameras is actually created from a smaller area than film. For digital sensors, the Crop Factor relates the ratio of the dimensions of this imaging area in cameras compared to a reference format. In effect, it shows a smaller area for an image constructed from the same lens but smaller sensor. That's why it's called a crop factor – implying that the lens' image is cropped in comparison to a 35mm film frame by a given factor. Because of this crop, the effective field of view (FOV) is reduced by a factor proportional to the ratio between the smaller sensor size and the 35mm reference film format.

A general rule in cameras to be remembered is that multiplying a lens' focal length by a camera's factor gives the focal length of a lens which, when